



Course: Introduction to Data Science
Data Science Project
April 12, 2026

1. Project Overview

In this project, you will perform a complete **data analysis pipeline** on a real-world dataset using Python (Pandas, NumPy, Matplotlib). The goal is to develop fundamental skills in **data cleaning, exploration, visualization, and interpretation**.

You are expected not only to analyze data, but also to formulate meaningful questions, extract insights, and communicate your findings clearly.

2. Group Work

- Groups of **three (3) students** (up to two groups of four (4) or one group of 2, if needed).
 - Each group selects **one dataset from the provided list**.
-

3. Datasets

1. Student Performance

Description: This dataset contains information about students' academic habits, background, and performance.

Features:

- **age:** Age of the student (in years).
- **study_hours:** Average number of hours spent studying per day.
- **attendance_rate:** Proportion of attended classes (between 0 and 1).
- **parent_education:** Highest education level of parents (categorical).
- **internet_access:** Whether the student has internet access at home.
- **sleep_hours:** Average number of hours of sleep per night.
- **stress_level:** Self-reported stress level (scale from 1 to 10).
- **engagement_score:** A synthetic measure of classroom engagement.
- **final_grade:** Final exam score (0–100 scale).

- **extra_notes**: Optional qualitative teacher notes.

2. Supermarket Sales

Description: This dataset simulates transaction records from a supermarket, including product and customer information.

Features:

- **branch**: Store branch identifier.
 - **category**: Product category purchased.
 - **unit_price**: Price per unit of the product.
 - **quantity**: Number of units purchased.
 - **customer_type**: Type of customer (e.g., member or non-member).
 - **payment**: Payment method used.
 - **hour**: Hour of the transaction (0–23).
 - **discount**: Discount applied to the transaction.
 - **total**: Total transaction amount.
-

3. Heart Health Dataset

Description: This dataset includes health indicators and lifestyle information used to assess the risk of heart disease.

Features:

- **age**: Age of the patient.
 - **gender**: Gender of the patient.
 - **cholesterol**: Cholesterol level (mg/dL).
 - **blood_pressure**: Resting blood pressure.
 - **heart_rate**: Resting heart rate (beats per minute).
 - **smoker**: Whether the patient is a smoker.
 - **exercise_level**: Physical activity level (Low, Medium, High).
 - **biomarker**: Experimental medical measurement.
 - **disease_risk**: Indicator of heart disease risk (0 = low, 1 = high).
-

4. Air Quality Dataset

Description: This dataset contains environmental measurements used to assess air pollution levels.

Features:

- **temperature**: Ambient temperature (°C).
- **humidity**: Relative humidity (%).
- **CO**: Carbon monoxide concentration.

- **NO2**: Nitrogen dioxide concentration.
 - **PM2_5**: Fine particulate matter concentration.
 - **wind_speed**: Wind speed.
 - **day_type**: Type of day (weekday or weekend).
 - **sensor_reading**: Aggregate sensor measurement.
 - **AQI**: Air Quality Index (higher = worse air quality).
 - **comment**: Optional qualitative air quality label.
-

5. Online Shopping Dataset

Description: This dataset represents user behavior on an e-commerce platform.

Features:

- **age**: Age of the customer.
 - **gender**: Gender of the customer.
 - **time_on_site**: Time spent on the website (minutes).
 - **pages_viewed**: Number of pages viewed during the session.
 - **device**: Device used (e.g., mobile or desktop).
 - **purchase_amount**: Total amount spent.
 - **discount_used**: Whether a discount was applied.
 - **click_rate**: Number of clicks or interactions.
 - **return_customer**: Whether the customer is returning (1) or new (0).
-

4. Project Tasks

Using **Pandas**, **NumPy**, and **Matplotlib**, your objective is to carry out a comprehensive **Exploratory Data Analysis (EDA)**, including the following steps:

1. **Data Understanding**
 - a. Examine the structure, features, and overall characteristics of the dataset.
2. **Data Cleaning**
 - a. Handle missing values.
 - b. Identify and remove duplicate records.
 - c. Resolve inconsistencies.
 - d. Detect and smooth noisy data.
 - e. Correct incorrect data types.
 - f. Identify outliers.
3. **Perform Univariate and Bivariate Analysis**
 - a. Compute summary statistics.
 - b. Visualize data distributions.
 - c. Analyze relationships between variables.
 - d. Evaluate correlations.
4. **Data Visualization**

- a. Create meaningful plots to support analysis. Plots must include a proper title, and clearly labeled x-axis and y-axis.
 5. **Interpret and Conclude**
 - a. Summarize findings and draw insightful conclusions.
-

5. Deliverables

1. An archive **.zip** file containing:
 - a. Jupyter Notebook (**.ipynb**)
 - i. Must run without errors
 - ii. Include code, explanations in markdowns, plots.
 - iii. Well-commented
 - b. Dataset (**.csv**)
 2. **Presentation**
 - a. Each group must prepare a brief presentation summarizing their work.
 - b. Present their work within 5 – 10 minutes, in front of the class.
-

6. Evaluation

#	Component	Weight
1	Data Cleaning	20%
2	Univariate and Bivariate Analysis	20%
3	Data Visualization	10%
4	Interpretation	20%
5	Code Quality	10%
6	Presentation	20%

Best of Luck